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Research paper

Using PV inverters for voltage support at night can lower grid costs

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ABSTRACT

Areas with sparse transmission lines are common in regions with high solar energy potential and need voltage support. This may require installing expensive voltage compensators, such as static synchronous compensators (STATCOMs). This expense can increase the cost and decrease the acceptance of large-scale adoption of solar power. Unlike current photovoltaic (PV) inverter controllers, which provide voltage support only during the day, commercially available augmented voltage controllers can provide voltage support at night. We examine whether PV inverters improve nighttime voltage on the grid and how much such an operation would cost compared to a STATCOM. We ran grid contingency analyses on a model for West Texas within the Electric Reliability Council of Texas (ERCOT) jurisdiction under spring and summer conditions to determine if PV inverters can support nighttime voltage under varying reactive power demand. The cost of reactive power has not been defined previously, especially in the context of the United States. Our methods and application provide a way to determine the cost of reactive power for both PV project developers and system planners. Allowing PV inverters to provide reactive power can reduce system costs by millions of dollars, or 4–15 times less costly than installing a STATCOM. We determined inverter voltage support costs by calculating the cost of earlier inverter replacements due to increased reactive power output and voltage controllers. The net system savings argue for ERCOT changing their voltage support policies to incentivize PV plants to provide voltage support at night.

1. Introduction

Loosely connected transmission systems are susceptible to voltage excursions due to the system's inability to meet their reactive power demand (Obadina and Berg, 1990). Voltage support is provided by reactive power, the result of a difference in phase between current and voltage, stated in units of volt–ampere reactive (VAR). Reactive power is needed for generators, motors, and transformers, and to compensate for reactive power losses on transmission lines. Typically, expensive voltage compensators are installed to maintain grid stability and system voltage at proper levels, especially in areas such as West Texas where transmission is sparse and voltage support is needed 24 h a day (ERCOT, 2019). Many regions with excellent utility-scale solar resources have similar issues. Inverter-based resources, mainly solar, provide voltage support during the day, but not at night. To provide nighttime voltage support, the Electric Reliability Council of Texas (ERCOT) installed two voltage compensators in West Texas in 2018 to mitigate voltage disturbances at a cost of $70 million (ERCOT, 2019). These add to the system costs of utility-scale solar power. Here we examine if a less expensive alternative is feasible and, if so, what policy changes are warranted.

Inverters that employ power electronics are used to convert DC power produced by photovoltaic (PV) solar panels to AC power for use on the grid when the sun is shining. When a PV plant is online, its inverters can provide voltage support (through the output of reactive power if it has appropriate electronics) to the grid (Loutan et al., 2017). However, at night inverters are no longer used. Hence, as things now stand, these assets sit idle for approximately 50% of their lifetime, and the voltage support PV plants provide during the day is not available at night.

Flexible alternating current transmission system (FACTS) devices provide dynamic voltage support. A common FACTS device is a static synchronous compensator (STATCOM). A STATCOM uses a voltage source converter to convert DC power to AC reactive power. Depending on the size and environmental conditions, STATCOMs can cost between $30 million and $100 million (California ISO, 2019; Weldy and Kawakami, 2018), depending on the equipment necessary for installation and interconnection, and location. STATCOM costs are typically given per unit of reactive power they provide, USD/kVAR, ranging from $77 - $290/kVAR.

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Costs for 18 STACOMs in 4 USA regional transmission organizations (RTOs) can be found in section S1 of the supplemental information (SI).

Varma et al. (2015) have shown that PV inverters with an augmented voltage controller can be used as a STATCOM at night. PV providing reactive power at night has been successfully field-tested in East Sussex UK by National Grid and Lightsource BP (Ambrose, 2019). Varma et al. (2019) argue that using a group of PV inverters for voltage support is about 50 times less costly than installing a FACTS device. That work performs a first-order capital cost comparison of STATCOM costs to those of an inverter supplying reactive power but does not include other costs incurred when an inverter provides reactive power support. Gandhi et al. (2019) quantified a reactive power provision in terms of the levelized cost of electricity for a solar plant. However, the authors do not model the dynamic response of PV inverters, nor discuss other sources of reactive power in the system and they imply that the PV inverters are the main sources of reactive power to serve load. A dynamic reactive device, such as a generator or FACTS device, is designed to respond quickly to voltage disturbances, not to serve base reactive load. Thus, a FACTS device must operate near zero reactive power output during normal operations to respond with its full reactive capability during a voltage deviation. Consequently, an improvement on the Gandhi et al. model should represent the expected dynamic response of a PV plant when providing reactive power.

Additional power flow through the inverter causes additional thermal stress on the electronic components of the inverter, particularly on the DC-link capacitor and power switching devices (transistors and diodes) (Yang et al., 2011). Previous work has not included the additional wear and tear on inverters, leading to a shorter lifespan and thus higher total inverter cost.

Methods to determine the cost of reactive power from a power plant on the bulk electric power system is missing from the literature, particularly in the context of the United States. In this paper we are the first to quantify the cost of reactive power from a solar PV power plant. We make the first comparison of the total cost of utilizing a PV inverter as a substitute for a STATCOM to that of a STATCOM, including wear and tear on the inverter and necessary equipment. An augmented voltage controller on the PV plants controller is necessary to operate the PV inverter at night and will need to be replaced during the lifetime of the PV plant. Using a model of the ERCOT grid, we examined whether PV inverters can prevent nighttime voltage excursions and used a discounted cash flow model to perform the cost comparison. Our proposed model can be used to estimate the cost of providing reactive power to the bulk power system from a solar PV plant. Using PV inverters in place of STATCOMs may provide additional sources of revenue for PV plant owners and remove one barrier to integrating renewables in loosely connected transmission systems.

Current policy within ERCOT requires that, above a certain MW threshold, generators must be able to provide voltage support to the grid. Intermittent renewable resources (IRR) are required to provide voltage support when operating at or above 10% of their rated maximum active power output, including low-voltage ride-through (LVRT) capability. IRRs do not have a voltage support requirement below that threshold (ERCOT, 2020a). However, inverter-based resources with an augmented voltage controller can provide voltage support at an output of zero active power (Loutan et al., 2017). We examined whether current policies in ERCOT should be updated to facilitate nighttime use of PV inverters, without altering the LVRT capability of the inverters.

2. Methods and data

To assess the feasibility and cost of using PV inverters for voltage support at night, we ran a power systems voltage analysis of an ERCOT model, simulated a grid-connected PV generator (in MATLAB/Simulink) to compute the lifetime of an inverter, and did a cost comparison between inverters that do and do not provide voltage support at night. We ran a power flow and contingency analysis on a synthetic ERCOT model to find low voltage grid conditions that can be resolved using existing PV infrastructure. We used two different system conditions for this model, spring and summer nighttime wind and load profiles to capture low load, high wind, and high load, low wind conditions. We ran the MATLAB/Simulink model to evaluate how reactive power causes the inverter to degrade over time. The inverter model had solar radiation and weather data with a thirty-minute temporal resolution, and we varied reactive power output to simulate a dynamic response of the inverter. We used weather and radiation data from the years 2000–2019 to account for variation in yearly weather. We then calculated the lifetime cost of an inverter that does not produce reactive power at night and compared that to the lifetime cost of an inverter that does produce reactive power at night.

2.1. Power systems model and power flow analysis

We conducted a power flow analysis to determine the feasibility of solar plants to mitigate a voltage event during nighttime conditions using the Texas A&M 2000 bus synthetic ERCOT model (Birchfield et al., 2017). The open-source Texas A&M 2000 bus synthetic ERCOT model was built in 2016 and matches closely the 2016 zonal load and generation levels in ERCOT as verified by load and generation data published on ERCOT's website (ERCOT, 2021a, 2017). A significant increase since 2016 in installed capacity of wind and solar generation, as well as increased load, has increased the need for better voltage control on the grid. Using publicly available data found on ERCOT's website, as well as data from the US Energy Information Administration (EIA), we added and updated wind and solar generation to the case to match 2019 penetration levels (EIA, 2020; ERCOT, 2020b). Additionally, we scaled load, specifically in West Texas, to match ERCOT 2019 load levels (ERCOT, 2020c). We modeled two different cases to capture two different nighttime conditions, Spring (lower load and higher wind), and Summer (higher load and lower wind).

We included only generators that have an apparent power rating of at least twenty mega volt–amperes (MVA) since only these are required to provide voltage support (ERCOT, 2020a). Both ERCOT and the EIA provide only MW levels. We estimated MVAR and MVA based on the 0.95 power factor requirement outlined in the ERCOT Nodal Protocols (ERCOT, 2020a). Therefore, each generator can produce reactive power at approximately one-third of its maximum active power output. Total apparent power is defined by $S = \sqrt{P^2 + Q^2}$, where P and Q are real and reactive power, respectively. Theoretically, an inverter can provide both real and reactive power up to its rated MVA limit if one of the variables is zero. We considered an inverter's ability to provide reactive power at night when active power is zero. However, we limited each plant's reactive output to two-thirds of its MVA rating when measured at the point it interconnects to the transmission grid because there are losses, mainly the filter and transformer, between the inverter AC terminals and the point of interconnect; limiting the reactive power output at the point of interconnection prevents overloading the inverter.

To determine the effect of a PV inverter improving the voltage profile of the grid, we ran power flow and contingency analysis to find key contingencies that cause low voltages using PowerWorld.

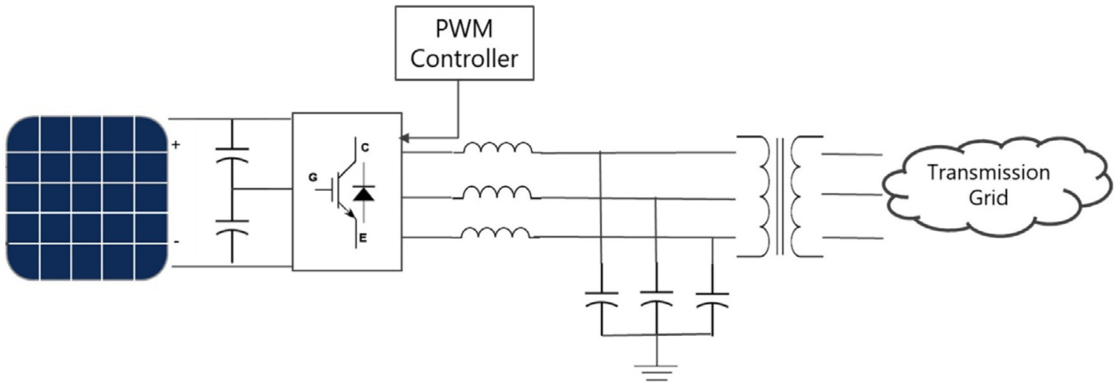

Fig. 1. Simplified schematic of the three-phase 2.8 MW PV power plant used in this model.

Voltage limits of this case are 0.9 per-unit (p.u.) to 1.1 p.u. (Per-unit voltage is the ratio of the actual bus voltage to its rated nominal voltage, where 1.0 p.u. is equal to the nominal rated value.) Post-contingency voltages in ERCOT must be within 0.9 p.u. to 1.0 p.u. (ERCOT, 2021b). Hence high, and low voltages need to be mitigated. Reactive power does not travel far across a transmission line, fifty to one-hundred miles. Therefore, only voltage deviations near an existing PV plant can be mitigated by reactive power produced at that plant. We then ran each case with a contingency with and without PV plants providing reactive power. Results are considered successful if voltage deviations near PV plants post-contingency can be mitigated by the inclusion of PV reactive power.

Table 1
PV plant and inverter parameters for MATLAB/Simulink model. The PV array is from the MATLAB/Simulink library and uses SunPower SPR-X20-445-COM modules.

Parameter	Value	Units
Rated Active Power Output	2.8	MW
PV Modules in Series per String	16	
Parallel PV Strings	394	
V_{dc}	1200	Volts
V_{ac}	625	Volts
Rated Apparent Power	3	MVA

2.2. Photovoltaic inverter model

To assess the effect of reactive power on an inverter, we modeled a three-phase, 2.8 MW grid-tied solar plant in MATLAB/Simulink (Fig. 1). Solar irradiance and temperature data were retrieved from the National Solar Radiation Database, which has half-hour temporal resolution (NREL, 2019). The simulator interpolated over each time step to capture voltage excursions, which ran using 1/3rd second-time steps. Using time-steps does not emulate real-time conditions and adds some uncertainty to the model. While the 20-cycle resolution will capture most varying reactive power demand, this may lead slightly different results in the inverter lifetime. Table 1 contains the PV plant and inverter parameters used in the model. The PV array is from the MATLAB/Simulink library and uses SunPower SPR-X20-445-COM modules. The model uses a three-level, neutral clamped insulated-gate bipolar transistor (IGBT) bridge with pulse width modulation (PWM) control. The PWM controller uses a Perturb and Observe maximum power point tracking (MPPT) algorithm to determine the DC voltage to achieve maximum power output from the PV array. That desired voltage value is used in the controller's voltage regulator, which uses a proportional integral (PI) controller to control the DC voltage. The current regulator has both a PI controller and feedforward to control AC active and reactive current using the Park/Clarke transformation. The reactive power demand ranged from ±2 MVAR, sampled from a normal distribution with a mean of zero. This was chosen to be two-thirds of the total MVA rating of the inverter, 3 MVA, to avoid overloading the inverter due to losses in the system.

2.3. PV inverter lifetime model

The main causes for inverter failures are power switching devices and the DC-link capacitor (Yang et al., 2011). Increased reactive power injection induces more thermal stress on inverter components, and capacitors are found to age faster due to power cycling and thermal stress (Gandhi et al., 2019). Thus, we assumed that the lifetime of the DC-link capacitor determines the lifetime of the inverter. Typical DC-link capacitors in power electronics are aluminum electrolytic capacitors and metalized polypropylene film capacitors (Wang and Blaabjerg, 2014). Electrolytic capacitors are no longer frequently used in inverters because of their short lifetime (Barker, 2020). Accordingly, we used a film capacitor for the inverter lifetime model. Two primary factors that influence the lifetime of a capacitor are its DC voltage and core temperature (Wang and Blaabjerg, 2014). Fig. 2 is a graphical depiction of the inverter lifetime model. The equations that estimate the operating life are S1 through S5 in section S3 of the SI.

We calculated the inverter lifetime when it is producing only active power by controlling reactive power to 0 MVAR at every time step. We then varied the reactive power loading to determine the effect that reactive power has on the lifetime. Comparing the two lifetimes gives the expected lifetime reduction of the inverter. The lifetime reduction is calculated as:

$$LR = L_{base} - L_{Q} \tag{1}$$

in which LR is the lifetime reduction of the inverter in years, L_{base} is the lifetime of the inverter that is only producing active power, and L_{Q} is the lifetime of the inverter that is producing real and reactive power.

We also calculated the inverter lifetime when it is producing reactive power during the day (L_{day}). By substituting L_{day} for L_{base} in Eq. (1), we calculated the expected lifetime reduction when an inverter produces reactive power during the day and night compared to an inverter producing reactive power during the daytime.

We modeled the Kemet Aluminum Can Power Film Capacitor (C44UHGT7210M87K) for the lifetime model. Table 2 gives characteristics of the capacitor taken from the datasheet and the model design (Kemet Electronics Corporation, 2020).

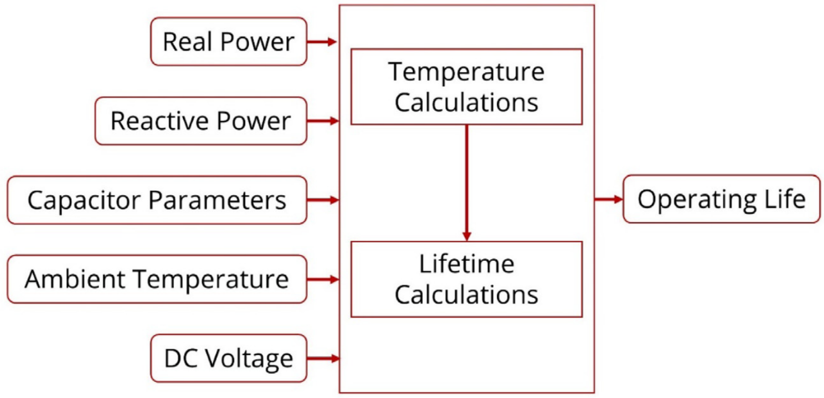

Fig. 2. Graphical depiction of the inverter lifetime model, which is estimated to be the same as the DC link capacitor. The two main factors that influence the lifetime are the DC voltage and core temperature. The DC voltage is a measured value from the simulation. The core temperature is influenced by the power flow and ambient temperature.

Table 2
DC capacitor link characteristics for inverter lifetime model.

Characteristic	Value	Units
Voltage Rating	600	Volts
Useful Life at 85 °C	10,000	Hours
Capacitance	2100 μF	Microfarads
Series Resistance (R_s)	1.7 mΩ	Milliohms
Thermal Resistance (R_{th})	3.6	°C/Watt
Voltage stress component (n)	7	
Capacitors in Series	2	
Capacitors in Parallel	15	

2.4. Cost of reactive power

We used a discounted cash flow model to find costs of a PV inverter that does not provide reactive power, provides reactive power during the day, and provides reactive power during the day and night. Inputs to the cash flow model for an inverter that does not provide reactive power or just during the day include an initial capital cost of the inverter at year 0 and an inverter replacement at year L_{base} or year L_{day}, respectively. Inputs to the cash flow model for the inverter that provides reactive power at night include initial capital costs of the inverter and controller at year 0, a controller replacement at the lifetime of the controller, an inverter replacement at year L_Q, and any multiple of L_Q if the inverter is replaced more than once. Eqs. (2) and (3) give respective costs of an inverter not providing reactive power, providing reactive power during the day, and providing reactive power during the day and night. C_{base} can be either the cost of an inverter providing reactive power during the day or not providing reactive power, depending on the inverter lifetime used.

$$C_{base} = \sum_{n=0}^{N} \frac{INV_{c,n}}{(1-r)^n} \tag{2}$$

$$C_Q = \sum_{n=0}^{N} \frac{INV_{qc,n} + CTRL_n}{(1-r)^n} \tag{3}$$

in which $INV_{c,n}$ is the cost of the inverter at year n, r is the discount rate, N is the lifetime of the PV plant in years, $INV_{qc,n}$ is the cost of the inverter providing reactive power at year n, C_Q is the cost of an inverter providing reactive power during daytime and nighttime, and $CTRL_n$ is the cost of the voltage controller at year n.

Combining Eqs. (2) and (3) yields the cost of reactive power:

$$C_Q - C_{base} = \sum_{n=0}^{N} \frac{INV_{qc,n} - INV_{c,n} + CTRL_n}{(1-r)^n} \tag{4}$$

To find the normalized cost for the inverter per unit of reactive power, we divided the total cost of providing reactive power by the inverter's reactive power capacity, in kVAR. Normalizing the cost to USD/kVAR allows the inverter cost to be compared directly with the cost of a voltage compensator. The USD/kVAR for an inverter is given by:

$$C_{VAR} = \frac{\sum_{n=0}^{N} \frac{INV_{qc,n} - INV_{c,n} + CTRL_n}{(1-r)^n}}{Q_{cap}} \tag{5}$$

where Q_{cap} is the reactive power capacity of the inverter.

3. Results and discussion

3.1. PV inverter behavior during nighttime operations

Allowing PV plants to provide reactive power when they are not producing active power improves post-contingency voltages of the grid. Fig. 3 is a histogram of bus voltages of the spring case after a contingency in which a single transmission line in West Texas has been removed. Each bar gives the number of busses within the specified voltage range. The orange bars show system voltages without PV plant reactive support. The blue bars show system voltages with PV plants providing reactive power. When PV plants are not included, there were twelve busses with voltages below 0.9 p.u. (the ERCOT limit). When PV plants provide reactive power, all twelve busses were brought back within normal limits because inverters were able to supply reactive power to the grid. All busses with voltages outside of the ERCOT limit were within 100 miles of a PV plant. The summer case showed results with fewer busses outside of the specified limits when PV plants supplied reactive support than when PV plants did not provide reactive support (Section S4 in the SI).

3.2. PV inverter lifetime reduction due to reactive power production

We ran a MATLAB/Simulink simulation using irradiance and temperature data from a location in West Texas that has high solar radiation (30.97°N, 103.3°W, near five large PV installations approximately 25 miles west of Ft. Stockton, Texas). Assuming every year will have the same operating conditions adds a level of unwarranted certainty to the inverter lifetime because

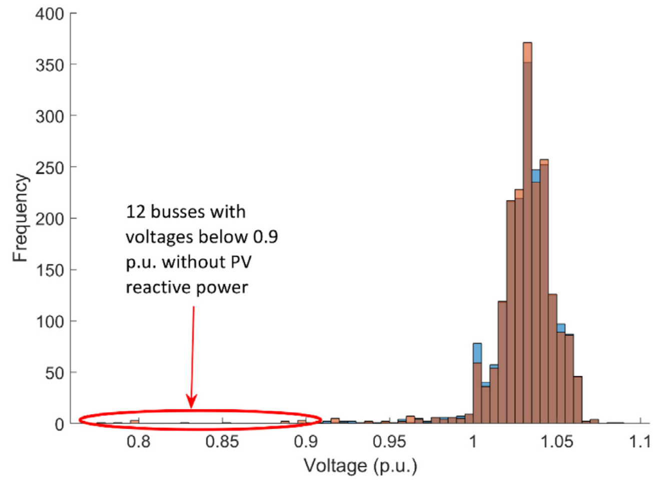

Fig. 3. A histogram of the distribution of system bus voltages in the 2000-bus model from the Spring case after a contingency in which a single line in West Texas has been removed. Each bar gives the number of busses within the specified p.u. voltage range. There are 12 voltages below 0.9 p.u. in the system that did not use PV inverters for voltage support (orange), compared to none in the system that did use PV inverters for voltage support (blue). Results for the Summer case are in Fig. S.1 found in section S4 in the SI.. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

weather conditions vary year to year. Therefore, we simulated each year from 2000 to 2019. For each of the twenty years, we simulated the MATLAB/Simulink inverter model with no reactive power loading, incremental reactive power loading from 4%–100%, where 100% is 2 MVAR, and at the expected reactive power loading of an inverter providing reactive power during the day and both day and night. The expected reactive power loading was estimated by simulating the operation of a PV plant to that of a STATCOM. A STATCOM responds to all voltage deviations that are observed on its control bus. Schemes are in place to keep the STATCOM output as close to zero as possible, so the STATCOMs can respond to fast voltage disturbances. STATCOMs that are in West Texas likely respond to 200 or more voltage events a year. We simulated a similar operation on the inverter by randomly varying the reactive power demand, both consuming and producing, at every time point, centered around 0 MVAR, which produced a reactive power loading of 11% when providing reactive power during the day and 25% when also providing reactive power at night.

We used results from each simulation to calculate the inverter's lifetime under each yearly and reactive loading conditions. Inverter lifetimes for each year can be found in section S5 of the SI. When the inverter is not providing reactive power, the average lifetime was found to be 25.9 $\pm$ 0.1 years. Varying the reactive power output of the inverter, we derived its lifetime based on its reactive power loading. The average lifetime of an inverter providing reactive power during the day was found to be 24.6 $\pm$ 0.1 years. By randomly varying reactive power loading of the inverter in the simulation during both the day and night, the model produced an average lifetime of inverter producing reactive power of 23.8 $\pm$ 0.1. For our 2.8 MW inverter model, Fig. 4. Number of years the lifetime of an inverter is reduced due to the amount of reactive power loading, where 100% is equal to 2 MVAR (two-thirds of its total power rating, 3 MVA). shows how many years the lifetime of the inverter is reduced as a function of reactive power loading. To obtain the average lifetimes under different operating conditions, we ran a stochastic simulation with results from the simulations to develop distributions of the base lifetime, day and night expected lifetimes, and incremental reactive loading lifetimes, and each was approximately normally distributed. We then took 1000 samples from each distribution and ran a Monte Carlo analysis to determine the lifetime reduction of the inverter under various loading conditions.

Table 3
Triangle distribution parameters for Monte Carlo analysis on discounted cash flow model.

	Median	Minimum	Maximum
PV Plant Costs	\$1.44/W	\$1.00/W	\$3.00/W
Inverter Costs	\$0.045/W	\$0.04/W	\$0.05/W
Controller Costs	1%	0.5%	1.5%

To summarize, an inverter that produces reactive power during the day and night has lifetime reductions of approximately two years and one year compared to inverters that do not provide reactive power and provide reactive power during the day, respectively. Temporal resolution and spatial factors do not significantly change these results and are discussed in more detail in section S5.1 in the SI.

3.3. Cost of reactive rower for a 2.8 MW inverter in West Texas

Using a discount rate of 6%, we ran a Monte Carlo simulation with the discounted cash flow model to estimate the present value of an inverter that does not provide reactive power and an inverter that does provide reactive power at night. We made the following assumptions for the present value for the inverter calculations. The lifetime of a PV plant is 30–40 years (Camargo, 2020). The lifetime of a PV-STATCOM controller is approximately 15 years (Rosenberger, 2020); therefore, we model a capital investment at year 0 and a replacement cost at year 15. The cost of the PV-STATCOM controller is approximately 1% of the PV plant capital cost (Varma et al., 2015). The median capital cost of a utility-scale solar plant is \$1.44/W, with the 20th and 80th percentiles being \$1.30/W and \$2.46/W, respectively (Bolinger et al., 2020). The cost of a utility-scale inverter is \$0.04 - \$0.05/W (Barker, 2020). We assumed that PV plant capital costs, inverter costs, and controller costs followed a triangular distribution. Table 3 outlines parameters for each distribution. We also ran a parametric analysis on the discount rate and controller lifetime to determine the effects of those variables on the results.

The mean present value for a 3 MVA inverter that does not provide reactive power is \$165,600 $\pm$ \$300. When the inverter provides reactive power during the day, the mean present value is \$167,700 $\pm$ \$500. The mean present value for the inverter that provides reactive power at night is \$248,300 $\pm$ \$1,500. Therefore, costs for a 3 MVA inverter are \$80,600 $\pm$ \$1,000 more when it provides reactive power at night than only during the day, corresponding to approximately a 48% cost increase (Fig. 5). The inverter provides $\pm$2 MVAR, a range of 4 MVAR. The normalized cost of an inverter providing reactive power is, assuming reactive power loading is what is produced by the model, \$20.20 $\pm$ \$0.40/kVAR. We note that an inverter is only 2%–4% of a PV generating plant's total capital costs, so an increase of 48% in the inverter cost does not substantially increase the cost of a PV project.

Sensitivity analysis results, varying the discount rate and voltage controller lifetime, are in Fig. 6.

4. Conclusions and policy implications

As more renewable generation is installed on the transmission grid, voltage support and stability in loosely connected transmission grids are becoming increasingly important. If sufficient

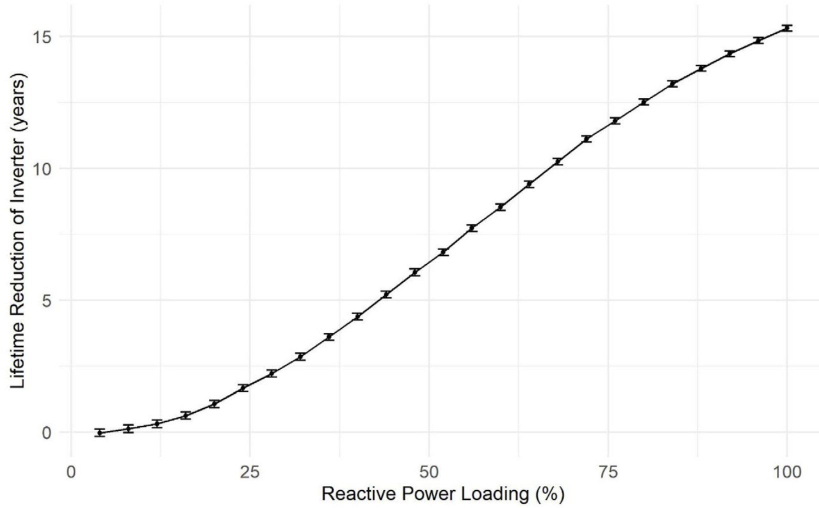

Fig. 4. Number of years the lifetime of an inverter is reduced due to the amount of reactive power loading, where 100% is equal to 2 MVAR (two-thirds of its total power rating, 3 MVA).

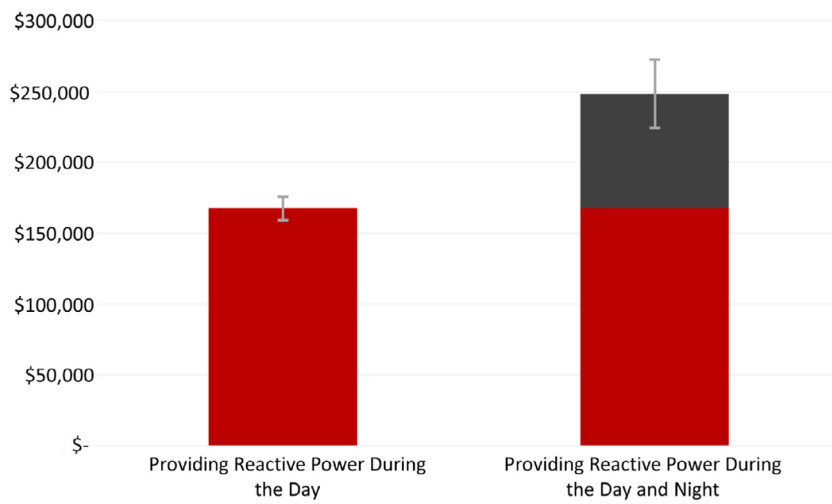

Fig. 5. Present values for 3 MVA inverters that do not and do provide nighttime reactive power to the grid. The bar on the right is at the expected reactive power production, 25% reactive power loading over its entire lifetime.

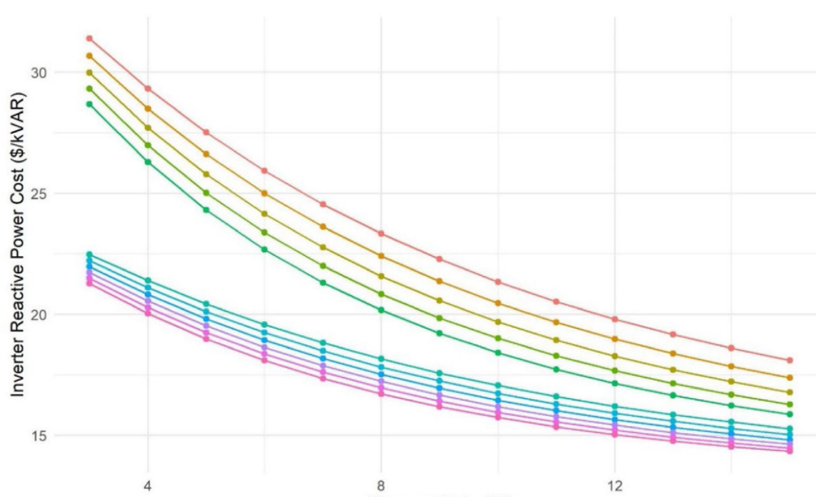

Fig. 6. Sensitivity plot of the discount rate and voltage controller lifetime. The voltage controller lifetime varies from 10 years (top line) to 20 years (bottom line), with one-year increments. We assumed that if the voltage controller's lifetime is less than 15 years, it is replaced twice and once if the lifetime is 15 years or longer. That difference causes the gap in inverter reactive power cost between years 14 and 15. As the discount rate increases, the cost of reactive power at night decreases.

voltage support is absent, a typical solution is to install an expensive FACTS device to provide voltage support, which can cost between \$77 and \$290/kVAR for a STATCOM. PV generation currently provides voltage support to the transmission grid during the day, but not at night. Furthermore, there are no policies in place at ERCOT that incentivize generators to provide grid support when they are not supplying active power.

PV plants have the capability of providing voltage support twenty-four hours a day using their inverters if they have appropriate voltage controllers. If a grid operator uses a PV plant for voltage support instead of installing a STATCOM, system costs can be reduced. Using a synthetic ERCOT model, under both spring and summer conditions, grid conditions are improved when PV plants provided reactive power. Twelve low voltage busses within 100 miles of PV plants were brought back within normal limits under spring nighttime conditions.

There are costs associated with PV plants providing reactive power. Inverter components wear out more quickly when providing reactive power, and there is a cost for an augmented voltage controller to allow inverters to provide voltage support at night. PV generators that control voltage during the day already expect a reduced lifetime of approximately one year. When a PV inverter provides additional reactive power at night, its lifetime is expected to decrease by an additional year, increasing inverter costs by approximately 48%, which is due to earlier replacements as well as voltage controller costs. However, an inverter is only 2%–4% of a PV generating plant's total costs, so an increase of 48% in the inverter cost does not substantially increase the cost of a PV project.

To assess system savings, the cost of a PV providing reactive power must be directly compared to that of a voltage compensator. We found that the average cost of an inverter providing reactive power is, $\$20.20 \pm \$0.40$/kVAR. Directly comparing that to STATCOM costs, a PV inverter is \$56 - \$269/kVAR less expensive. This difference has a large range due to the large range in STATCOM costs, which are due the STATCOM size, environmental conditions of the location, and the equipment necessary for installation and interconnection. Varma et al. (2015) reported that PV-STATCOM costs can be up to 50 times less than that of an equivalent STATCOM, which accounted for an augmented voltage controller. Our analysis showed that operating PV inverters at night is 4 to 14 times less costly. The cost difference is due to a shorter lifespan of PV inverters as well as a more detailed discounted cash flow model that accounts for controller replacements. Therefore, if a 50 MVAR STATCOM is found to be needed at a grid location where an equivalently sized or larger PV plant exists, system savings can range from \$2.8 million to \$13.5 million when using the average cost of an inverter providing reactive power. There have been at least five STATCOMs installed on the ERCOT grid in the past twenty years, with two in West Texas in 2018 (ERCOT, 2019, 2003; Oskoui et al., 2006). As load and PV penetration increases on the grid, more voltage support may be required to meet system demands. Having a PV generator that can provide voltage support 24-hours a day gives system planners more flexibility in choosing voltage compensators.

The cost of reactive power is based on the reduced lifetime of the inverter along with the lifetime of the inverter and the inverter controller replacements relative to its reactive power capability. Different PV penetration levels may lead to differing reactive power demand. Adding more PV inverters to the system with voltage support at night functionality will reduce reactive power demand on each inverter, reducing the cost of reactive power on each inverter. However, installing more PV inverters without voltage support at night functionality may increase reactive power demand on inverters with the proposed functionality, reducing the lifetime of the inverter and increasing reactive power costs. System operators and planners should carefully consider the expected reactive power demand when determining reactive power costs.

4.1. Policy implications

Considering the large potential cost savings for avoiding the installation of a voltage compensator, ERCOT should consider updating its policies to allow generating plants to provide voltage support even when they are not providing active power. ERCOT has policies in place that require generating plants larger than 20 MVA to provide voltage support to the grid when a generator is above a certain threshold. ERCOT can revise their Nodal Protocols to require generators to provide voltage support service when they are not providing active power. However, that will likely not be favorable to generator owners because they will incur additional costs. A different approach is a market solution. ERCOT can also consider revising its protocols to include a location-dependent ancillary service that creates a market for reactive power. For locations that need reactive power, nearby PV plants can offer expected reactive power capacity on an hourly basis. Because PV plants are already required to provide reactive support during the day, this solution can be implemented only at night.

Operating PV at night should also be considered in other independent system operators (ISO), such as the California ISO (CAISO). CAISO conducted a low generation reactive power test to assess the feasibility of PV plants providing reactive power support when the power plant is offline. The results showed that the plant is capable of producing and consuming reactive power at nearly zero active power output (Loutan and Gevorgian, 2017). At night and during off-peak hours, CAISO experiences high voltages that may be mitigated by PV plants providing reactive power (Loutan, 2021). However, the PV plants will not provide that support without policies and incentives. Thus, CAISO should consider updating its policies to include a location-dependent ancillary service for reactive power.

4.2. Technical and policy hurdles

There are some drawbacks of this proposed operation. Existing PV plants may be unable to augment their voltage controllers to provide this service due to hardware design limitations. Unless the functionality can easily be updated by a firmware change on some plants, this service might be for only new generators in the interconnect queue. However, that may not be a serious issue as there were 81 GW of PV projects in the ERCOT interconnection queue as of late 2020 (ERCOT, 2020d). Furthermore, voltage control in ERCOT is managed by transmission operators, using transformer tap changers, inductor and capacitor banks, and FACTS devices. If a FACTS device is replaced by a PV inverter, transmission operators no longer have control over its operation. Outage and voltage coordination issues can be resolved by increased coordination and communication with ERCOT, transmission operators, and generator operators. To ensure redundancy, it may be desirable to provide incentives to overbuild the number of inverters. There are normally several large inverters at a single plant. Therefore, additional needs for redundancy may be mitigated by rotating outages of inverters at a PV plant if there is still adequate reactive supply within available inverters. Otherwise, a smaller FACTS device may be needed, which will still lead to system savings. During the day, the full reactive power capability of the inverter may not be available. While solar PV power plants are required to meet the specified 0.95 power factor requirements, the system may require additional reactive power support. In this case, it may also be desirable to provide incentives to overbuild the number of inverters or install a smaller FACTS device, still leading to system savings.

4.3. Additional research

Other electronics in the inverter will also be affected by an increased flow of reactive power. Both transistors and diodes will be subjected to thermal stress due to reactive power flow (Callegari et al., 2019). Therefore, the lifetime of those components will also affect the lifetime of an inverter. Further analysis on the effect of reactive power on transistors and diodes will provide further insight into how reactive power reduces the lifetime of an inverter and will give more insight to generator owners on how the proposed operation will affect their equipment.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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